Perturbation Self-Consistency: A Cheap Black-Box Baseline for Predicting Reasoning Robustness, and Where It Fails

Alan De Bortolo
Design Master Solutions LLC
info@designmastersolutions.com

Abstract

The AIMO Interpretability Challenge asks whether a math-reasoning model's answer to a problem survives perturbations that leave the deductive chain intact. We submit a training-free, black-box method that measures this property directly at test time: the evaluated checkpoint answers the original problem and three rule-based perturbations (letter-swap typos, an irrelevant distractor sentence, character renaming) with greedy decoding, and the case is called robust only if every answer agrees. The method needs no labels, no weights access beyond generation and no learned readout, and it runs inside the 3600 s budget with full coverage. On the validation leaderboard it reaches 0.619 accuracy on the Main Model Track (submission 1019467) and 0.571 on the Small Model Track (submission 1019144), against 0.500 for the always-`False` predictor on the Small track (submission 1019672). We report this as a partly negative result: the gain over the majority class is modest, the always-`False` predictor reaches 0.68 on the public sample, and the public sample associates robustness with original-problem accuracy, which a single greedy answer does not measure. The 28 rows represent only 24 distinct model–problem pairs: weighting pairs equally lowers the always-false baseline from 0.679 to 0.625. We release code, tests and the ablation bundles.

1 Introduction

The challenge label marks a problem–model pair as robust when the model's accuracy holds across every perturbation applied by the organizers and non-robust when at least one perturbation clearly degrades it. The exact criterion is not published. Most interpretability methods would learn a proxy for this label from internal signals. We take the opposite route as a baseline: if the label is about behaviour under perturbation, measure that behaviour on the evaluated model itself, with perturbations we can generate offline. This gives a reference point for how much of the task is solvable without any access to internals, and its failure modes point at what a white-box method has to add.

2 Method

Perturbations. For each problem we build up to three variants that never touch mathematical tokens (digits, LaTeX commands, operators): (i) *typos*: swap two inner letters in every fourth plain word of length ≥ 5; (ii) *distract*: append one fixed irrelevant sentence with a number in it; (iii) *rename*: replace common character names when they appear (no-op otherwise). Organizer perturbations that need a generator model (domain transfer, rephrasing, expert edits) are not reproduced.

Answers. The original and each variant are answered by the evaluated checkpoint with greedy decoding, batched and sorted by length, loaded offline with `local_files_only`. The final answer

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is the last `\boxed{}` after the reasoning, else the last number after `</think>`; truncated reasoning yields no answer.

Decision. A case is robust iff the original produced an answer and every variant's answer equals it. No answer, a time-out or a load failure yields `False`, the majority class in every public sample (robust share 0.32–0.375).

Budget. The 3600 s limit covers all models and problems. A global budget of 3300 s is split evenly across the remaining `are_robust` calls and passed to `generate(max_time=...)` per batch, so every call returns and coverage stays at 1.0.

3 Experimental setup

Main Track models: Qwen3.5-4B (thinking), Skywork-OR1-Math-7B, Olmo-3-7B-Think, DeepSeek-R1-0528-Qwen3-8B and gpt-oss-120b (low/medium/high effort); the Small Model Track drops gpt-oss-120b. We submitted the full method to both tracks and, on the Small track, three ablations built from the same file by changing one constant (`PC_PERTURBATIONS`): typos only, distractor only, and `none` (always `False`).

4 Results

Track	Variant	Codabench ID	Accuracy	Coverage
Small	full (typo+distract+rename)	1019144	0.571	1.00
Main	full (typo+distract+rename)	1019467	0.619	1.00
Small	always `False`	1019672	0.500	1.00
Small	typos only	1020199	0.571	1.00
Small	distractor only	1020200	0.571	1.00
Small	typos + 2 sampled originals	1020807	0.571	1.00
Main	typos + 2 sampled originals	1020808	0.524	0.76

Table 1: Validation leaderboard results, read from the Codabench API on 29–30 October 2026.

Ablations. Each single perturbation reaches the same Small-track accuracy as the full method (0.571 for typos only and for distractor only). Equal aggregate accuracy does not establish that the methods flag the same cases. Without paired predictions we cannot attribute the signal to a particular edit or test whether the methods have complementary errors. We observed no aggregate accuracy gain from the combined edits; this is not evidence of equivalent predictions.

5 Analysis of the public sample

The 28-row public sample (seven frontier models, AIMO-2025 reference problems) carries the per-case statistics behind the label. Two observations shape how this method should be read.

Base accuracy is a near-necessary condition. None of the 11 rows with base accuracy below 1.0 is robust; 9 of the 17 rows with base accuracy 1.0 are. After collapsing repeated model–problem pairs these counts are 0/7 and 9/17. This is a descriptive association, not the labeling rule: the organizers do not publish the exact threshold for clear degradation. Our method compares a single greedy answer per prompt, so it cannot see this: a model that is right 6 times out of 10 and greedy-stable across our three edits is called robust and is wrong.

Model-level association is confounded by problem selection. Label means per model range from 0.0 (gpt-oss-120b, Gemini, Qwen3-8B) to 1.0 (GPT-5.2, GLM-5.1). These groups contain only one to seven rows and cover different problem subsets, so the sample cannot establish whether model identity or problem selection explains the difference. A sampling-variance prior is a hypothesis to test, not a conclusion from these group means.

Implication, tested. The natural extension adds a sampled estimate of base reliability on the original problem (temperature 1.0, top-k 40, top-p 0.95, as in the organizers' generation) and requires both self-agreement and stability under perturbation. We tested the cheapest version at the same compute as the full method (four generations per case: greedy original, typo variant, two sampled originals; knob `PC_SAMPLES`). On the Small track it reaches 0.571 (submission 1020807), the same as the greedy-only method. The two-sample variant therefore showed no aggregate accuracy improvement on this validation set. Without paired predictions or a per-case reliability measure, this tie does not establish whether the variants distinguish reliable from occasionally-right models or make the same errors. On the Main track it drops to 0.524 with 0.76 coverage (submission 1020808, five invalid predictions): sampled generations on the larger checkpoint run longer than greedy ones and push cases past the 3600 s budget, so the variant costs accuracy where the base method had full coverage. This run exposes a coverage problem for this implementation and budget, not an upper bound on black-box methods. Shorter decoding, adaptive allocation or another sampling policy remain untested.

6 Conclusion

Perturbation self-consistency is a cheap, label-free baseline that runs within budget on every evaluated model, and on the validation leaderboard it beats the always-`False` predictor by 7 points on the Small track (0.571 against 0.500), with 0.619 on the Main track. The margin is small. The public sample suggests base-problem reliability as a possible missing signal, but repeated pairs and confounded model/problem coverage prevent a causal explanation of leaderboard errors. We offer the method and this negative result as a calibration point for black-box approaches.

Reproducibility

Code (MIT): `https://github.com/DesignMaster-Solutions/aimo-perturb-consistency`. The submission bundle is `solution.py` alone (plus an empty `small.txt` for the Small track); ablation bundles are produced by `./ablate.sh typo distract none`. No training data and no training codebase are used. Tests: `uvx pytest -q tests`. Public-sample counts are reproduced by `report/analysis/public_sample.py`; its JSON output records the exact dataset URL, revision and SHA-256. This analysis is descriptive and is not used to train or tune the submitted predictor.

AI usage statement

The method, code, tests and this report were developed with the help of an AI coding assistant (Claude; Codex for the public-sample audit and report corrections) under the author's direction; all numbers are read from the Codabench API and the public sample.

Limitations

Our perturbation set covers three of the organizers' seven perturbation families. Greedy decoding differs from the sampled generation used to build the labels. The validation leaderboard has few cases per track, so differences of one or two cases move accuracy by several points. Public-sample rows are not independent: 28 rows cover eight problems and 24 distinct model–problem pairs. No rows are exact duplicates, but four repeat a pair; pair-weighted always-false accuracy is 15/24 (0.625), versus 19/28 (0.679) row-weighted. The robust labels agree within each repeated pair. We report both weightings as a sensitivity analysis, not a correction to the official score. These labels must not be confused with held-out leaderboard predictions; no significance or generalization claim follows from them.